

OPOLE, Poland

WMO: 125300

Lat: 50.6269N

Lon: 17.9689E

Elev: 535

StdP: 14.41

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.4	12.3	1.6	6.1	7.6	6.6	7.9	13.7	20.5	37.4	18.4	36.9	3.2	90	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	88.1	68.6	84.3	67.1	81.1	65.7	70.5	83.2	69.0	80.6	67.5	77.8	6.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	99.6	74.3	65.1	94.8	72.8	63.8	90.3	71.6	34.7	83.7	33.4	80.7	32.2	77.6	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.4	12.6	DB	1.8	93.5	7.2	2.9	-3.3	95.6	-7.5	97.3	-11.6	99.0	-16.8	101.1	(4)
(5)			WB	1.3	74.0	7.1	2.3	-3.8	75.6	-8.0	76.9	-12.0	78.2	-17.1	79.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	30.5	33.3	39.5	49.3	57.5	63.6	67.4	66.8	58.2	49.7	41.3	33.3	(6)	
	DBStd	14.96	9.48	9.21	7.71	7.65	6.71	6.20	5.68	5.64	6.21	7.48	7.63	8.65	(7)	
	HDD50	2425	606	469	334	103	12	0	0	0	6	100	275	520	(8)	
	HDD65	6043	1070	889	792	474	248	98	38	45	217	476	712	984	(9)	
	CDD50	2162	0	1	7	81	243	410	541	522	253	90	13	1	(10)	
	CDD65	303	0	0	0	1	14	58	114	102	13	1	0	0	(11)	
	CDH74	3160	0	0	0	37	201	620	1160	988	152	2	0	0	(12)	
	CDH80	948	0	0	0	3	29	177	399	310	30	0	0	0	(13)	
(14) Wind	WSAvg	5.7	6.5	6.5	6.7	6.2	5.5	5.1	4.9	4.6	4.9	5.4	6.0	6.5	(14)	
(15) Precipitation	PrecAvg	22.8	1.3	1.1	1.5	1.5	2.4	2.9	3.6	2.2	2.2	1.5	1.5	1.3	(15)	
	PrecMax	28.4	2.2	2.8	3.3	2.5	4.5	6.6	10.9	4.3	4.1	3.5	2.9	3.9	(16)	
	PrecMin	14.8	0.4	0.1	0.5	0.2	0.9	0.7	0.3	0.5	0.6	0.2	0.0	0.5	(17)	
	PrecStd	3.1	0.5	0.6	0.7	0.6	0.9	1.4	2.4	0.9	1.1	0.8	0.7	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	56.2	65.2	78.7	83.5	90.1	93.1	92.9	84.7	73.5	62.1	54.1	(19)	
		MCWB	46.9	48.3	51.9	60.6	64.0	70.6	68.7	70.1	65.1	61.2	54.1	48.8	(20)	
	2%	DB	48.0	51.8	59.9	72.4	79.5	85.1	88.6	87.1	78.2	68.8	58.8	50.1	(21)	
		MCWB	44.4	45.9	49.4	56.6	63.7	68.2	68.3	68.5	63.2	59.0	52.5	46.1	(22)	
	5%	DB	44.9	48.8	55.5	67.7	75.5	80.8	84.4	83.1	74.2	65.1	55.6	46.6	(23)	
		MCWB	41.6	44.1	47.4	54.1	61.5	65.6	67.1	67.0	61.8	57.6	50.6	43.6	(24)	
	10%	DB	42.4	45.6	51.6	63.7	71.4	77.0	80.8	79.5	70.4	61.2	52.5	43.8	(25)	
		MCWB	39.6	41.6	45.3	52.4	59.4	64.1	66.0	65.5	60.4	55.4	48.6	41.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	49.4	53.3	61.7	67.4	72.3	72.4	72.1	67.2	62.4	55.5	49.3	(27)	
		MCDB	50.9	54.0	63.0	74.8	78.7	85.7	85.8	85.5	78.6	70.3	60.8	53.4	(28)	
	2%	WB	44.5	46.8	50.5	58.4	64.7	69.6	70.6	70.0	65.1	60.1	53.1	46.4	(29)	
		MCDB	47.6	51.3	57.9	70.1	75.7	82.0	82.8	82.9	74.8	67.1	57.9	49.8	(30)	
	5%	WB	42.0	44.3	48.1	55.6	62.8	67.4	69.1	68.4	63.3	58.1	51.0	43.8	(31)	
		MCDB	44.5	48.2	54.3	65.0	73.0	77.7	80.1	80.1	71.6	63.9	54.9	46.3	(32)	
	10%	WB	39.8	41.8	45.8	53.3	60.7	65.5	67.5	66.9	61.5	55.9	48.8	41.4	(33)	
		MCDB	42.3	45.4	51.1	61.9	69.5	74.5	77.5	76.5	68.5	60.5	52.1	43.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	11.3	14.7	18.9	19.3	18.8	19.4	20.1	17.5	14.4	10.4	8.9	(35)	
		MCDBR	10.8	15.5	22.4	26.8	26.7	26.3	27.2	26.7	24.9	20.3	15.2	10.9	(36)	
	5% WB	MCWBR	8.3	10.7	13.6	13.5	12.6	11.4	10.1	10.5	11.6	11.8	10.0	8.4	(37)	
		MCWBR	10.5	14.4	20.7	23.4	23.9	23.5	23.0	23.2	20.9	18.7	13.8	10.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.318	0.345	0.384	0.436	0.429	0.427	0.448	0.437	0.402	0.374	0.348	0.314	(40)	
		taud	2.303	2.279	2.219	2.129	2.196	2.238	2.191	2.228	2.294	2.336	2.353	2.350	(41)	
	Ebn at Noon		221	246	256	255	263	264	256	252	247	232	208	205	(42)	
		Edh at Noon	22	29	36	44	43	42	43	40	34	27	21	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		257	487	833	1297	1586	1671	1590	1426	1015	595	290	196	(44)	
	RadStd		47	82	125	168	187	165	192	124	145	86	40	24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.92	N/A	N/A	+1.44	N/A	+0.68	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	+0.86	N/A	N/A	+1.31	N/A	N/A	N/A	N/A	+106	+40

Nomenclature: See separate page